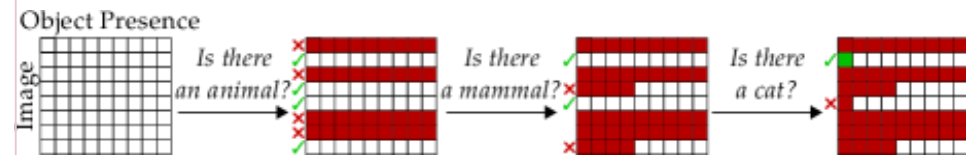


# Representative Visual Forms of World-Model Datasets

Images, video hours, episodes, trajectories, and benchmark tasks encode fundamentally different learning signals

## (a) Static visual semantics



### ImageNet-1K

1.28M images | category labels

**Rich object semantics, no temporal order, no action, no interaction**

Original source: ImageNet / ILSVRC paper figure

## (b) Human-object action video



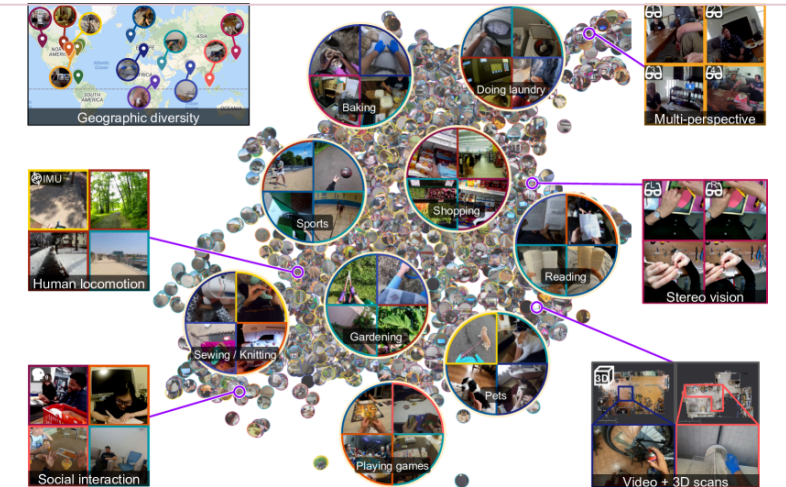
### Something-Something-v2

220K videos | temporal action labels

**Temporal order, fine-grained actions, lacks executable robot control**

Original source: Goyal et al., ICCV 2017, Figure 1

## (c) Egocentric interaction



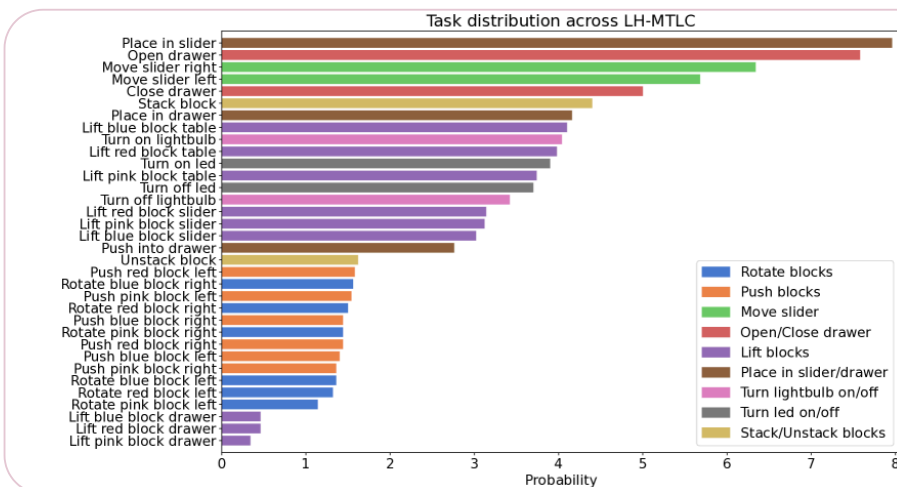
### Ego4D / EPIC-KITCHENS

3,670 h / 55 h | egocentric interaction

**First-person view, hand-object affordance, anticipation**

Original source: Ego4D paper, Figure 1

## (d) Simulation control



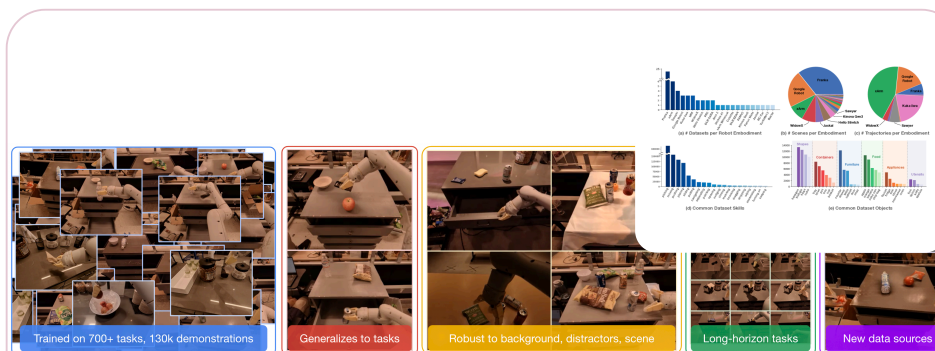
### CALVIN / MuJoCo

simulation | state-action-reward

**Controllable physics, dense state/action access, synthetic domain**

Original source: CALVIN paper, Figure 1

## (e) Real-robot demonstration



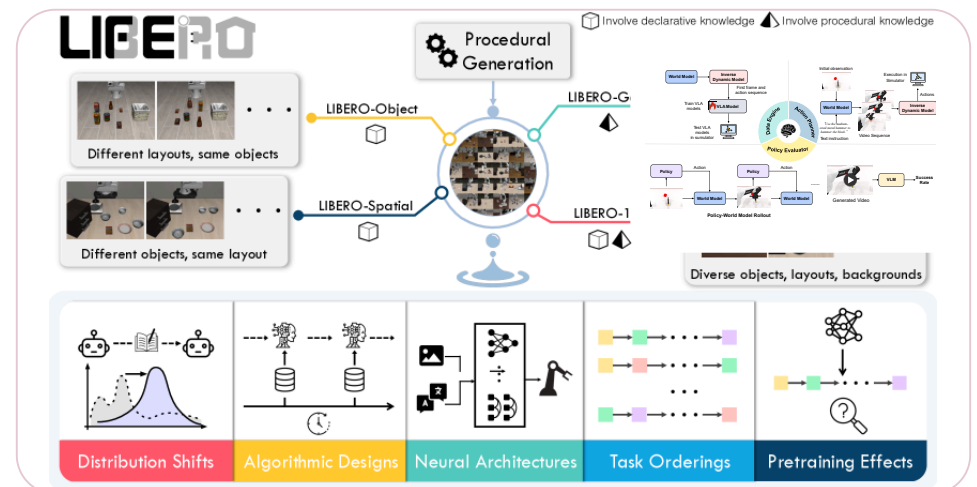
### RT-1 / Open X-Embodiment

130K episodes / 1M+ trajectories | robot actions

**Real action supervision, language grounding, cross-embodiment**

Original sources: RT-1 teaser; Open X-Embodiment Figure 1 inset

## (f) Embodied benchmark



### LIBERO / WorldArena

130 tasks / functional evaluation | closed-loop utility

**Success, robustness, functional utility, interactive evaluation**

Original sources: LIBERO task suites; WorldArena evaluation framework inset